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AI adoption rate and corporate green innovation efficiency: Evidence from Chinese energy companies

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A R T I C L E I N F O

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A B S T R A C T

The advent of artificial intelligence (AI) technology has led to transformative shifts in the human landscape. Moreover, as a potent driving force behind the evolution of energy companies, green innovation has the potential to be supported by advanced technologies such as AI. However, academic research exploring the association between the AI adoption rate and green innovation in energy companies is scarce. This study analyzes data on Chinese listed energy companies from 2010 to 2020. Our findings indicate that energy companies with extensive AI adoption exhibit higher green innovation efficiency. This finding is particularly pronounced among firms that report substantial participation in environmental, social, and governance activities. However, our findings reveal that executives who focus on short-term benefits can undermine the positive influence of AI adoption on green innovation. These main findings are notably significant for energy companies where the roles of chief executive officer and board director are unified, state-owned enterprises, and companies that do not hold bank shares. This study offers novel insights and valuable guidance for policymakers regarding the strategic development of energy companies, thereby bridging a significant gap in the literature.

1. Introduction

The Chinese economy has recently undergone swift and significant expansion, and while this has led to a multitude of impressive achievements, it has also given rise to numerous challenges (Song et al., 2018; Wang et al., 2022). Prior studies have highlighted that the steady evolution of listed companies is intricately linked to both the advancement of the national economy and societal progress (Hodson, 1984; Becker et al., 2016; Ren et al., 2022) and to the provision of foundational assurance concerning the long-term sustainable growth of the companies themselves (Khan et al., 2021). Because they play a pivotal role in socioeconomic progress, energy companies not only transform natural resources into indispensable daily commodities but also shoulder the critical obligations of energy conservation and emissions reduction (Liu et al., 2021). In response to China's call for a circular economy and the transition from traditional to green and environmentally friendly industries, energy companies are increasingly directing their investments toward the creation of innovative green products (Guo and Zhang, 2023; Liu et al., 2023; Song et al., 2023; Wang et al., 2024). From the innovation theory perspective, the creation of premium green products requires the support of cutting-edge technology (Dou and Choi, 2021; Tang et al., 2023), including artificial intelligence (AI). Additionally, the application of AI could provide unprecedented benefits and opportunities for companies in terms of innovation across sectors (Akter et al., 2021; Maliha et al., 2021). However, researchers have largely overlooked the role of AI technology in the green innovation efficiency (GIE) of energy companies. To overcome this gap in the literature and extend the existing research stream in this field, this study provides empirical findings on the relationship between AI adoption rates and corporate GIE among Chinese listed energy companies.

Corporate GIE refers to the capability to manufacture the greenest and most innovative products at minimum cost (Asimakopoulos et al., 2020). It paves the way for lowering research and development (R&D) costs and bolstering corporate productivity (Porter and van der Linde, 1995; Piao et al., 2022) while also serving as a yardstick for global environmental policy (Orsatti et al., 2020). Moreover, companies that prioritize green innovation can reap the benefits of achieving their sustainable development objectives, strengthening their competitive advantage (Chen et al., 2006), generating more employment opportunities (Kunapatarawong and Martínez-Ros, 2016), and retaining high-caliber talent (Fazal-e-Hasan et al., 2023). Most prior studies underscore the value and significance of green innovation and green patents or

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offer a broad interpretation of the catalytic effects of technological progression on green innovation. However, these studies do not examine whether the evolution of a specific technology can foster green innovation, resulting in a lack of research on the role of AI technology in relation to green innovation within energy companies.

Intelligent technology is poised to drastically reshape established corporate innovation models; thus, businesses must judiciously harness AI to enhance market competitiveness and drive progress (Sjödin et al., 2021). Through a comprehensive literature review spanning from 1966 to 2022, Bahoo et al. (2023) demonstrated that, in the era of information technology intelligence, executives should integrate AI technology into companies' core competitiveness rather than relegating it to the role of a mere auxiliary tool. Furthermore, the emergence of AI technology presents both opportunities and challenges for listed companies. For example, empirical evidence from Germany underscores how AI technology is utilized for both profit generation and process innovation, while the benefits of the associated technological revolution are particularly evident among German companies that have utilized AI for several years (Rammer et al., 2022). Proponents highlight the positive effects of AI technology on corporations; however, skeptics, as highlighted by Grashof and Kopka (2023), critically contend that AI may exert a detrimental influence on innovation efficiency alongside its benefits. Thus, companies must establish a more robust AI innovation management model to fully prepare for the challenges of using AI technology, including the aversion to AI replacing part of the workforce (Füller et al., 2022).

The intricate interplay between AI adoption rates and company outcomes has been the subject of scholarly exploration across various dimensions. Nevertheless, a discernible research gap remains, specifically with respect to empirical investigations addressing the effects of AI adoption rates on GIE, in the context of energy companies. The exigency of this empirical inquiry underscores its significance as a pressing and unresolved practical dilemma requiring scholarly attention. To identify the intricate effects of AI adoption rates on energy companies' innovation output, constructive recommendations for both energy companies and policymakers are needed. These suggestions must aim to address the existing challenges and propel future advancements in this dynamic landscape. For energy companies navigating the integration of AI technologies and GIE, we pose the following research question: What are the effects of the AI adoption rate on GIE within energy companies? How can the AI adoption rate be quantified to represent AI technology utilization within green companies more objectively and accurately? What are the mechanisms underlying this relationship?

To answer these questions, we apply the following research design. First, we download data on Chinese energy companies listed on the A-share markets of Shanghai and Shenzhen from 2010 to 2020. Second, we utilize textual analysis techniques based on machine learning to construct AI adoption metrics. The main advantage of textual analysis is its ability to objectively and accurately depict the AI adoption rates of energy companies (Pennebaker et al., 2003). Third, we propose three research hypotheses and empirically investigate the influence of AI adoption rates on GIE over the decade used as the sample period. Fourth, to elucidate the mechanisms underlying the relationship between the AI adoption rate and GIE in energy companies, we introduce corporate environmental, social, and governance (ESG) performance, and executive myopia as moderating effects. Finally, we conduct robustness checks and heterogeneity analyses to further validate our research findings.

We identify a positive correlation between the extent of energy companies' AI technology utilization and their GIE. This finding is statistically significant, suggesting that AI use enhances green innovation progress in energy companies. Our mechanism analysis reveals that energy companies' ESG performance can positively moderate the effects of AI adoption on corporate GIE. However, executive myopia negatively moderates this relationship. Thus, we encourage energy companies to increase their involvement in ESG activities while cautioning executives to prioritize long-term development over immediate benefits. Furthermore, our research suggests that the effects of AI adoption on energy companies' GIE exhibit heterogeneity depending on whether the company has a chief executive officer (CEO) concurrently serving as the board chairperson, is state-owned, or holds bank shares.

This study makes three contributions. First, our findings offer valuable insights for energy companies in the context of green innovation while also enriching the evidence in related fields. We suggest that energy companies adopt AI technology to aid in the development of green patents and products. This would not only benefit immediate product development but also contribute to companies' long-term sustainable growth. Moreover, exploring the mechanisms underlying ESG can hold additional reference value for energy companies. Along with the integration of AI technology, increasing ESG participation could augment the positive effects of AI use on green innovation.

Second, for corporate executives, an excessive focus on immediate performance is not advantageous, because executive myopia can diminish the positive effects of AI on energy companies' GIE. This finding provides a reminder for company executives to align their development strategies with long-term objectives, thereby offering valuable guidance for corporate leadership.

Third, the formulation of policies that encourage energy companies to apply AI technology represents a prudent strategy for government bodies and policymakers. This approach could not only enhance the green innovation of energy companies but also contribute to the national achievement of the United Nations' Sustainable Development Goals (SDGs). The global energy concerns are undeniable, and our primary objective is to guarantee universal access to advanced energy services while enhancing the efficiency and utilization of renewable sources. Furthermore, green innovation has the potential to expedite SDG attainment across nations and regions. AI technology serves as a catalyst to encourage energy corporations to increasingly develop green patents. Consequently, this study offers valuable insights for strategic planning within energy companies and actionable policy recommendations for decision-makers in the field.

The remainder of this paper is organized as follows. Section 2 reviews the related literature and proposes the hypotheses. Section 3 describes the research data and methodology. Section 4 interprets the results. Section 5 elaborates on the mechanisms and heterogeneity analyses. Finally, Section 6 presents conclusions and offers recommendations.

2. Literature review and hypothesis development

2.1. Corporate green innovation and AI adoption

In response to increasingly stringent environmental protection policies, energy companies have been progressively exploring new technologies to enhance their GIE (Rahman et al., 2023). Improving the efficiency of green innovative technologies not only allows energy companies to meet environmental protection standards more quickly but also helps reduce the cost of developing green patents and stimulate green product output (Witajewski-Baltvilks and Fischer, 2023; Li et al., 2024). Most previous studies on corporate green innovation have focused on multinational corporations, suggesting that the development of green technology is the primary source of value creation for such entities (Kim et al., 2021). Other studies have examined green innovation output and quality from the perspective of mandatory corporate social responsibility (CSR) disclosure methods (Mbanyele et al., 2022), considered the effects of greenhouse gas emissions and green innovation on corporate financial performance (Benkraiem et al., 2023), and assessed green innovation through the lens of environmental information transparency (Feng et al., 2023). Furthermore, researchers have examined the effects of human-induced extreme events on green innovation (Zheng et al., 2021) and the consequences of natural extreme events from a global perspective (Wei et al., 2023). However, the exploration of new technologies such as AI and their potential to drive GIE in energy companies remains relatively uncharted territory in academic research.

As AI continues to mature and proliferate across various industries and fields (Alekseeva et al., 2021), the potential of this cutting-edge technology has piqued the interest of energy companies. AI can benefit companies in numerous ways, including more efficient energy resource allocation (Melander and Pazirandeh, 2019), additional stock returns (Fotheringham and Wiles, 2023), increased productivity (Czarnitzki et al., 2023), and expanded corporate influence (Giacomoni, 2022). Previous studies suggest that companies in the transitional phase of AI integration experience a range of positive and negative effects. On the one hand, AI can bolster company profits, increase employment rates, and enhance operational efficiency. On the other hand, the short-term costs associated with AI implementation, such as larger employee salaries, can increase. Nevertheless, increased profits generally outweigh the costs of AI implementation (Mishra et al., 2022). Moreover, the multifaceted aspects of AI technology not only provide technical convenience for companies and organizations but also bolster their management capabilities, improve their competitiveness in complex markets, and create more value in business-to-business marketing (Mikalef et al., 2023). However, the rise of AI technology is not without its drawbacks. Lui et al. (2022) conducted a study of 62 listed companies investing in AI and found that investment in and the application of AI technology can negatively impact company value, particularly for non-manufacturing companies and companies with weak information technology capabilities. Their study also suggests that investors tend to be skeptical about companies that heavily promote the use of AI in investment announcements.

Companies are increasingly adopting AI technology to achieve their development goals and enhance their business operations. For energy companies, the primary challenge lies in improving green technology innovation, which serves as a fundamental indicator of their sustainable development and crucial criterion for meeting the requirements of governments' green environmental protection policies. Therefore, whether energy companies' adoption of AI technology can positively affect their GIE warrants further exploration. Notably, a gap exists in the literature regarding the influence of the AI adoption rate on green innovation within energy companies. To address this research gap and enrich the relevant literature, we propose the following hypothesis:

H1. : The AI adoption rate is positively correlated with corporate GIE in energy companies.

2.2. Role of corporate ESG activities in AI and green innovation

In recent years, the criteria for evaluating companies have evolved from focusing on traditional economic performance to more comprehensively considering ESG indicators. ESG indices typically encompass information on aspects such as environmental protection, social responsibility, governance structure, employee benefits, supply chain management, and financial performance. The aim is to showcase the sustainable development performance of enterprises transparently to investors, stakeholders, and the public (Apergis et al., 2022). Thus, ESG reports have emerged as crucial tools for assessing CSR and sustainable development (Wen et al., 2022). Prior studies have found that the disclosure of ESG data can positively impact a company's sustainable development performance, making it essential to incorporate ESG into a company's comprehensive evaluation (Drempetic et al., 2020). Furthermore, the extent of ESG policy implementation within companies can stimulate innovation capabilities and encourage greater commitment to research on new technologies (Broadstock et al., 2020), such as introducing AI technologies and stimulating corporate GIE. Moreover, studies have identified a positive relationship between ESG and green finance, suggesting that government departments should establish a more systematic and robust ESG disclosure system to monitor corporate performance in green finance (Zhang, 2023). From an agency perspective, agents with a strong preference for ESG activities are more inclined to invest in a company's tangible and intangible green assets such as green products and green patents. While they may receive lower returns by pursuing such an approach, they can achieve higher utility from green assets (Pástor et al., 2021).

The insights gleaned from the abovementioned literature make it apparent that enterprises' ESG ratings have become an indispensable metric. ESG not only has economic significance at the corporate level but also influences various other aspects, including green finance, green assets, sustainable development, and green innovation. The above literature review reveals that engaging in ESG activities can motivate companies to adopt AI technologies and foster increased green innovation. Moreover, the application of AI technologies may have a considerable impact on corporate GIE. However, a conspicuous gap exists in the literature concerning the effects of integrating ESG activities with AI adoption rates on GIE in energy companies. To enrich this literature stream and provide further support for research in this field, we propose the following hypothesis:

H2. : ESG activities positively moderate the effect of AI adoption rates on corporate GIE.

2.3. Executive myopia and green innovation

In addition to the macro-level analysis, we consider the effects of internal executives' personal characteristics on both AI adoption and GIE. As the decision-makers and managers within a company, executives' behavioral habits, psychological states, cognitive abilities, and management levels can influence a company's development and future planning (Mishra, 2014; Dong and Doukas, 2021; Aghamolla and Hashimoto, 2023). According to myopia theory, behaviors that prioritize short-term benefits at the expense of long-term development are shortsighted. Early studies posited that executive myopia could affect a company's operations and planning, thereby diminishing its development potential (Larwood and Whittaker, 1977). Scholars have noted that, although executive myopia can boost a company's short-term profits (Li, 2019), it can undermine the company's capacity for long-term sustainable development. This behavior can also erode a company's confidence in developing new products and inhibit its innovation abilities (Seo et al., 2020). Moreover, executives who fixate on short-term benefits often overlook potential challenges companies may encounter in their future development, which can impede sustainable development and further decelerate innovation. Conversely, executives who exhibit a long-term focus can maintain a certain competitive advantage in long-term enterprise development, stimulate innovation at all levels, and provide ample motivation for sustainable development (Antia et al., 2010).

Nevertheless, AI technology implementation requires both a substantial financial investment and patience, as it may not yield immediate benefits. This challenges executives' forbearance and demands that they exhibit a certain degree of long-term vision. The inclination to invest in AI technology varies among executives depending on their levels of myopia. The above analysis indicates that AI technology can significantly contribute to a company's green innovation and ultimately benefit the organization. Thus, executive myopia is a worthwhile concept to incorporate into this study to understand its role in the effects of AI adoption on GIE in energy companies. Therefore, we propose the following hypothesis:

H3. : Executive myopia negatively moderates the effect of the AI adoption rate on corporate GIE.

3. Data of listed energy companies and methodology

3.1. Selecting the sample of listed energy companies

Our initial research sample is drawn from the annual reports of Chinese companies listed on the Shanghai and Shenzhen stock exchanges from 2010 to 2020. The financial data are collected from the China Stock Market and Accounting Research (CSMAR) database, while regional-level data are collected from the National Bureau of Statistics. Subsequently, we refine our sample by excluding firms that do not operate within the energy sector, as defined in the *Industrial Classification of National Economy* documents issued by the State Administration for Market Regulation of the People's Republic of China. We also exclude firms with incomplete data for key indicators and all continuous variables are winsorized at the 1% level.

The final dataset comprises 1208 annual observations derived from 279 listed energy companies. To create an indicator of the rate of AI adoption for listed companies, we utilize the WinGo textual analytics database to execute precise word frequency searches of annual reports from 2010 to 2020. We then subject the search results to logarithmic processing to generate the AI adoption rates of the listed companies.

3.2. Applying textual analysis to measure the AI adoption rate

Building on the work of Li (2010), we employ textual analysis to identify a series of Chinese terms that encapsulate the concept of an "artificial intelligence horizon." Subsequently, we formulate an AI indicator using both a lexicographical approach and management discussion and analysis (MD&A) statements gleaned from the annual reports of 500 companies. The effectiveness of this indicator in capturing the AI adoption rate has been verified through a battery of tests including actual comparison, internal consistency reliability, analysis of variance, and economic consequence tests. Variance decomposition tests further demonstrate that this indicator does not merely reflect AI adoption driven by external pressures.

Given that speakers often employ a range of semantically similar terms to articulate identical thoughts, distinguishing between vocabulary similarities is crucial. Drawing on Word2Vec machine-learning technology, we represent the identified vocabulary words as multidimensional vectors and then determine the similarity between these vectors (Hong et al., 2022). We train the annual report corpus by leveraging the Continuous Bag of Words (CBOW) model in Word2Vec. This model can be represented as follows:

$$max\sum_{w\in C}logp(w|Context(w)) \tag{1}$$

where C symbolizes the corpus, w denotes the center word, and $Context(w)$ signifies the context of w. The CBOW model predicts the probability of a given word being used, based on its context. By maximizing Eq. (1), the Word2Vec word vector can be obtained and words similar to the center word can be identified by discerning vector similarity. Moreover, the CBOW model effectively circumvents the subjectivity inherent in manually defining word lists and the weak correlation typically associated with general synonym tools. In this study, we utilize the lexicon methodology to count the number of times terms related to "artificial intelligence" appear in MD&A statements. We then subtract the annual number of words from the previous year's word count to obtain the net number of words for each year. Finally, the rate of AI adoption is computed by dividing the net number of times AI-related words appear by the total word count. A higher value for this index implies more extensive application of AI in an energy company. The detailed seed words are provided in Appendix A.

3.3. Measure of corporate GIE

Inspired by the method of measuring innovation efficiency (Li et al., 2023), we utilize the ratio of green innovation output to total green innovation input to construct the GIE index. However, because the availability of total green innovation input data for listed companies is limited, we employ annual R&D expenditure as a proxy for total green innovation input. Additionally, we measure listed companies' green innovation output by the total number of green patents, utility models, and design patents as the natural logarithm plus one, as shown in Eq. (2). The GIE index represents a company's effectiveness in converting green input into green output. A higher GIE index value essentially signifies a company's increased effectiveness in green innovation production.

The formula we use to measure GIE is as follows:

$$GIE = \frac{ln[(green\ patents) + (utility\ models) + (design\ patents) + 1]}{Annual\ R\&D\ expenditure}. \tag{2}$$

3.4. Baseline model

Upon completing the tasks and testing Hypothesis 1 as outlined in the preceding sections, we construct a fixed-effects model to empirically examine the effects of the AI adoption rate on corporate GIE. The baseline model is expressed as follows:

$$GIE_{i,t} = \alpha + \beta_1 AI_{i,t} + \boldsymbol{\beta_2 Control_{i,t}} + \varphi_j + \tau_t + \epsilon_i, \tag{3}$$

where $GIE_{i,t}$ refers to the dependent variable that is the GIE of company i in year t; $AI_{i,t}$ indicates the independent variable that is the AI adoption rate of company i in year t; $\boldsymbol{Control_{i,t}}$ is a vector of control variables including corporate financial controls, corporate CEO controls, and environmental policy controls; φ_j represents industrial-fixed effects; and τ_t represents year-fixed effects.

We control for firm-, CEO-, and policy-level factors that may influence corporate GIE. The firm-level controls include total assets, leverage, return on assets, market-to-book ratio, Tobin's Q, shareholding ratios of the top 1 shareholders, firm age, and cash flow. CEO-level controls comprise the CEO's years of education and information technology skills. Policy-level control is an environmental regulation indicator in the city where the company is located. Table 1 provides comprehensive definitions of all variables used in this study.

4. Empirical results and analysis

4.1. Summary statistics

The summary statistics for the key variables used in this study are presented in Table 2. The main independent variable (i.e., AI adoption rate) exhibits a maximum value of 3.336 and a minimum value of −0.054, with a standard deviation of 0.254. This wide range underscores the considerable variability in the independent variable across the dataset. The primary dependent variable of interest is corporate GIE, which has a maximum value of 0.402, minimum value of 0, and standard deviation of 0.062. These results indicate that, although some energy companies have not produced any green innovation products, the highest ratio of green production to total innovation input is 40.2%. Thus, some companies have a maximum output of 40.2 green products out of every 100 products produced. Table 3 presents the correlation matrices are presented in Table 3.

4.2. Main results of the baseline model

The estimated results concerning the relationship between the AI adoption rate and corporate GIE, as depicted in Model (3), are shown in Table 4. Initially, we perform a simple regression without any controls or fixed effects, as shown in Column (1). Subsequently, we incorporate

Table 1
Variable definitions.

Variables	Definitions	Reference
Independent Variables		
AI	The artificial intelligence adoption rate of Chinese listed companies determined by textual analysis.	(Hong et al., 2022)
Dependent Variables		
GIE	The green innovation efficiency of Chinese listed companies measured by the ratio of green innovation production and total input in green innovation	(Li et al., 2023)
Firm-level controls		
Asset	Nature logarithm of the total assets	(Hu et al., 2023)
Leverage	Total liability divided by total assets	(Hu et al., 2023)
ROA	The ratio of total income to the total value of assets	(Hu et al., 2023)
BM	The market-to-book ratio is measured by book value/total market value	(Hu et al., 2023)
Tobin's Q	The value of Tobin's Q	(Hu et al., 2023)
TOP1	The shareholding ratios of the top 1 shareholder equals the ratio of number of shares of top 1 shareholder to the total shares of all shareholders.	(Hu et al., 2023)
Firm Age	The years of firms since they were established	(Hu et al., 2023)
Cash Flow	The ratio of cash flow generated by company's operating activity to total assets.	(Hu et al., 2023)
CEO-level controls		
CEO education	Number of years of CEO's education	(Amore et al., 2019)
CEO information	Dummy variable equal to 1 if CEOs have information technology background, 0 otherwise	(Amore et al., 2019)
Policy-level controls		
Environmental Policy	The level of environmental regulation policy in each year of cities where companies are located.	(Duan et al., 2021)
Moderating Variables		
ESG	The degree of corporate environment, social responsibility, and governance	(Wen et al., 2022)
Myopia	The degree of corporate executive myopia	(Antia et al., 2010)

This table explains the variables used in this paper including variable names and variable definitions, including reference.

firm-, CEO-, and policy-level controls as detailed in Column (2). Finally, Column (3) includes industry- and year-fixed variables. The coefficients across Columns (1)–(3) are uniformly positive and significant, suggesting that the use of AI technology can effectively increase corporate GIE, which provides evidence supporting Hypothesis 1. Thus, our main findings substantiate the assertion that the adoption of AI technology can serve as a catalyst for green innovation in energy companies, which has the potential to not only incentivize GIE but also amplify commitment to ecological sustainability. This offers valuable insights that can guide the trajectory of the national ecological economy. By corroborating and broadening the scope of the existing literature (Martínez-Ros and Kunapatarawong, 2019; Li et al., 2022; Ho et al., 2024), these primary conclusions contribute significantly to the ongoing discourse in this field.

Table 2
Summary statistics.

Variables	Observation	Mean	S.D.	Min	Max
AI	1208	0.053	0.254	−0.054	3.336
GIE	1208	0.044	0.062	0	0.402
Asset	1208	23.495	1.614	20.081	28.636
Leverage	1208	0.562	0.180	0.052	0.925
ROA	1208	0.027	0.047	−0.389	0.215
BM	1208	2.034	1.760	0.066	9.946
Tobin's Q	1208	1.431	0.784	0.799	11.422
TOP1	1208	0.411	0.159	0.086	0.758
Firm Age	1208	2.920	0.345	1.099	3.555
Cash Flow	1208	0.044	0.063	−0.174	0.241
CEO education	1208	17.698	1.575	14	23
CEO information	1208	0.025	0.156	0	1
Environmental Policy	1208	0.002	0.002	0.0001	0.023
ESG	1208	6.661	1.309	1	9
Myopia	1208	8.515	5.529	0	43

This table illustrates summary statistics of variables, including number of observations, mean value, standard deviation, minimum values, and maximum values.

4.3. Endogeneity

4.3.1. General difference-in-difference model

In 2017, President Xi articulated the following environmental protection concept: "Lucid waters and lush mountains are invaluable assets. It's everybody's duty to protect the environment well." Subsequently, Chinese governmental bodies and listed companies have increasingly focused on environmental conservation and sustainable development. For energy companies, the principal objective and key response to national environmental policies entail achieving strategic goals for green and sustainable development. Green innovation often serves as the factor that determines whether a company can meet its environmental requirements and attain its green and sustainable development objectives. A company with high green innovation output can consistently produce a steady stream of reliable green products that benefit society. However, further investigation is needed to determine whether the application of AI technology can help companies achieve these green objectives. Drawing on the methodology of Nunn and Qian (2011), 2017 is designated as the study's time point. A generalized difference-in-difference model is then used to investigate the relationship between the AI adoption rate and corporate GIE, as depicted in the following model:

$$GIE_{i,t} = \alpha + \beta_1 AI \times post17_{i,t} + \boldsymbol{\beta_2 Control_{i,t}} + \varphi_j + \tau_t + \epsilon_i, \tag{4}$$

where the primary focus is the interaction term of $AI \times post17_{i,t}$, which represents the effects of the AI adoption rate on corporate GIE post-2017. Both the sign and magnitude of the regression coefficient for this interaction term help to ascertain whether the influence of the AI adoption rate on corporate green innovation is positive or negative. In addition, the significance of the coefficient offers insight into the statistical evidence supporting the related results.

The results of Model (4) are shown in Table 5. The coefficient of the interaction term is 0.028 and significant at the 1% level. This implies that a government's environmental policy can bolster the impact of AI adoption on corporate green innovation to a certain extent. Thus, for government departments and policymakers, the implementation of suitable environmental policies and promotion of AI use within energy companies can stimulate the output of corporate green innovation, thereby achieving green sustainable development. These policy recommendations are consistent with those made in previous studies (Fabrizi et al., 2018; Huang et al., 2019; Song et al., 2021; Zhao et al., 2023).

4.3.2. One-year lag period

To mitigate the bias induced by reverse causality (Bernstein et al., 2017), and considering that the effects on corporate GIE may not be

Table 3
Pearson correlation coefficient.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1.GIE	1														
2.AI	0.123***	1													
3.Asset	0.265***	0.260***	1												
4.Leverage	−0.053*	0.025	0.400***	1											
5.ROA	0.099***	0.067**	0.028	−0.290***	1										
6.BM	0.030	0.075***	0.642***	0.575***	−0.184***	1									
7.Tobin's Q	−0.119***	−0.094***	−0.462***	−0.375***	0.067**	−0.462***	1								
8.TOP1	0.131***	0.148***	0.428***	0.007	0.154***	0.133***	−0.121***	1							
9.Firm Age	−0.091***	0.049*	−0.203***	0.033	−0.047*	0.015	−0.037	−0.234***	1						
10.Cash Flow	0.028	0.217***	0.185***	−0.176***	0.277***	−0.042	0.014	0.185***	0.018	1					
11.CEO education	0.279***	0.047	0.871***	0.268***	0.033	0.485***	−0.312***	0.437***	−0.295***	0.200***	1				
12.CEO information	−0.042	−0.006	0.026	0.039	−0.032	0.034	0.005	0.035	−0.003	0.014	0.041	1			
13.Policy	−0.109***	−0.027	−0.023	0.026	−0.037	−0.111***	0.095***	0.096***	−0.111***	0.015	−0.019	−0.066**	1		
14.ESG	0.102***	0.206***	0.412***	0.021	0.036	0.203***	−0.103***	0.211***	−0.021	0.176***	0.356***	−0.060**	0.015	1	
15.Myopia	−0.050*	−0.015	0.126***	0.032	−0.038	0.045	−0.036	0.126***	−0.114***	−0.030	0.140***	0.031	0.014	0.077***	1

This table shows the Pearson correlation coefficient of the key variables of main regression. * stands for 10% significant level, ** and *** stands for 5% and 1% level, respectively. Overall, most of the coefficients are statistically significantly correlated, which support us for further analysis in this study.

Table 4
Baseline regression results.

	(1)	(2)	(3)
	GIE	GIE	GIE
AI	0.030***	0.022***	0.034***
	(0.008)	(0.008)	(0.007)
Asset		0.008***	0.008***
		(0.003)	(0.003)
Leverage		−0.041***	−0.079***
		(0.012)	(0.013)
ROA		0.057*	−0.020
		(0.032)	(0.035)
BM		−0.007***	−0.005***
		(0.002)	(0.002)
Tobin's Q		−0.005***	−0.004***
		(0.002)	(0.002)
TOP1		−0.011	−0.007
		(0.011)	(0.012)
Firm Age		0.001	0.0004
		(0.006)	(0.006)
Cash Flow		−0.107***	−0.031
		(0.025)	(0.028)
CEO education		0.009***	0.008***
		(0.002)	(0.003)
CEO information		−0.019**	−0.033***
		(0.009)	(0.007)
Policy Regulation		−3.086***	−1.217*
		(0.730)	(0.735)
Constant	0.042***	−0.249***	−0.252***
	(0.002)	(0.057)	(0.051)
Observation	1208	1208	1208
R-Squared	0.015	0.150	0.235
Industry FE	No	No	YES
Year FE	No	No	YES

This table estimates the effects of corporate AI adoption rate on green innovation efficiency. The independent variable is AI adoption rate of listed energy companies in 2010 to 2020. The dependent variable is corporate green innovation efficiency. Firm level controls are natural logarithm of total asset (*Asset*), total liability divided by total assets (*Leverage*), ratio of total income to the total value of assets (*ROA*), market-to-book ratio (*BM*), value of Tobin's Q (*Tobin's Q*), shareholding ratios of the top 1 shareholders (*TOP1*), age of firms (*Firm Age*), ratio of cash flow (*Cash Flow*), CEO's years of education (*CEO education*), CEOs technology background (*CEO information*), level of environmental regulation policy (*Policy Regulation*). Column (1) has neither control nor fixed variables; Column (2) contains control but no fixed variables; Column (3) contains both control and fixed variables. The robust standard errors are at firm-level. * stands for 10% significant level, ** and *** stands for 5% and 1% level.

immediately observable following the adoption of AI technology, we introduce a one-year lag for the independent and control variables. As Table 6 shows, we initially run the model without the controls or fixed effects, as detailed in Column (1), before adding the controls in Column (2). Column (3) includes both the control variables and industry- and year-fixed effects. The results remain positive and significant across Columns (1)–(3), underscoring the robustness of our results and suggesting a low likelihood that problems will arise from reverse causality. Overall, the lagged method, which integrates independent and control variables with a one-year lag into the model, yields significant insights for energy companies. The incorporation of AI technology positively influences both current and future GIE. Therefore, from a long-term perspective, escalating R&D and investment in AI technology is advantageous for the sustained growth of energy companies. This highlights the strategic value of AI as a tool for promoting ecological sustainability and efficiency in the energy sector.

4.3.3. *Propensity score matching*

To address potential sample selection bias, we apply propensity score matching (PSM) (Rosenbaum and Rubin, 1983) to the data. We divide the AI adoption rate at the median, assigning values greater than the median to the treatment group (marked as 1), and values less than or equal to the median to the control group (marked as 0). We then conduct

Table 5
General difference-in-difference results.

	(1)
	GIE
AI × Post17	0.028***
	(0.007)
Asset	0.010***
	(0.003)
Leverage	−0.079***
	(0.013)
ROA	−0.024
	(0.035)
BM	−0.005***
	(0.002)
Tobin's Q	−0.004**
	(0.002)
TOP1	−0.005
	(0.012)
Firm Age	0.0004
	(0.006)
Cash Flow	−0.025
	(0.028)
CEO education	0.007***
	(0.003)
CEO information	−0.033***
	(0.007)
Policy Regulation	−1.142
	(0.758)
Constant	−0.271***
	(0.052)
Observation	1208
Adjusted R-Squared	0.231
Industry FE	YES
Year FE	YES

This table shows the difference-in-difference results of the effects of corporate AI adoption rate on green innovation efficiency by taking 2017 as cutting point. The main results are shown in column (1) and the interaction term of AI adoption rate and the dummy variable of year 2017 is the main focus. All control variables, and industry and year fixed effects are included. The robust standard errors are at firm-level. * stands for 10% significant level, ** and *** stands for 5% and 1% level, respectively.

Table 6
Lagged regression (1 year).

	(1)	(2)	(3)
	GIE	GIE	GIE
L.AI	0.038***	0.025***	0.038***
	(0.008)	(0.009)	(0.007)
Control Variables	NO	YES	YES
Observation	871	871	871
Adjusted R-squared	0.021	0.150	0.246
Industry FE	No	No	YES
Year FE	No	No	YES

In order to eliminate the bias caused by reverse causality, and considering that the effects may not be immediately observable, this table shows the lagged effects of corporate AI adoption rate on green innovation efficiency from 2010 to 2020. Column (1) has neither control nor fixed variables; Column (2) contains control but no fixed variables; Column (3) contains both control and fixed variables. The robust standard errors are at firm-level. * stands for 10% significant level, ** and *** stands for 5% and 1% level, respectively.

PSM using the same control variables and fixed effects as in Model (3). As shown in Table 7, the sign and significance of the PSM coefficients in Column (1) mirror the pattern observed in the main results and reinforce the robustness of the baseline model. These findings imply that the application of AI technology is positively and significantly associated with corporate GIE.

Table 7
Propensity score matching.

	(1)
	PSM
AI	0.037***
	(0.008)
Asset	0.012**
	(0.005)
Leverage	−0.084***
	(0.022)
ROA	−0.083
	(0.054)
BM	−0.001
	(0.002)
Tobin's Q	−0.002
	(0.003)
TOP1	0.011
	(0.017)
Firm Age	0.008
	(0.009)
Cash Flow	0.031
	(0.042)
CEO Education	0.003
	(0.004)
CEO information	−0.032**
	(0.013)
Policy	−2.154**
	(1.019)
Constant	−0.295***
	(0.077)
Observation	596
Adjusted R-squared	0.310
Industry FE	YES
Year FE	YES

This table contains the results of Propensity Score Matching (PSM) of the effects of corporate AI adoption rate on green innovation efficiency from 2010 to 2020. Column (1) shows the results of PSM test; The robust standard errors are at firm-level. * stands for 10% significant level, ** and *** stands for 5% and 1% level, respectively. To further address potential endogeneity resulting from variables omitted due to sample selection bias, we also perform a Heckman test (Heckman, 1979) on the listed energy companies. Specifically, we employ the Heckman command in STATA to regress the dependent variable, independent variable, and control variables, while also including the industry and year fixed effects. We use the two-step method to output the results, that is, the AI adoption rate is positively associated with corporate GIE, with the magnitude and standard error closely mirroring the results in Column (1) of PSM test. Furthermore, the inverse Mills ratio, as computed via the two-stage Heckman method, is not significant, suggesting that there is insufficient statistical evidence to indicate that the regression model is affected by sample selection bias. This lends further robustness to our findings.

4.4. Robustness checks

4.4.1. Narrow study period

According to a notice issued by the Ministry of Industry and Information Technology of China on the "*Industrial Green Development Plan (2016–2020)*," from 2016 to 2020, China's listed companies entered a new stage of promoting green development and stimulating green innovation. Inspired by Zhang (2021), the objective of this section is to verify the robustness of the impact of the AI adoption rate on corporate GIE from 2016 to 2020. As shown in Column (1) of Table 8, we first run a regression without any controls or fixed effects. Subsequently, we add the controls, as detailed in Column (2). Finally, the Column (3) incorporates the industry- and year-fixed effects. The results presented in Columns (1)–(3) affirm that corporate GIE is positively influenced by the

Table 8
Narrow studying period from 2016 to 2020.

	(1)	(2)	(3)
	GIE	GIE	GIE
AI	0.032***	0.027***	0.037***
	(0.008)	(0.008)	(0.007)
Asset		0.003	0.006
		(0.004)	(0.004)
Leverage		−0.020	−0.072***
		(0.014)	(0.016)
ROA		0.037	−0.056
		(0.041)	(0.044)
BM		−0.006***	−0.005**
		(0.002)	(0.002)
Tobin's Q		−0.008***	−0.006**
		(0.003)	(0.002)
TOP1		0.001	0.009
		(0.014)	(0.015)
Firm Age		0.004	0.008
		(0.008)	(0.008)
Cash Flow		−0.109***	−0.009
		(0.030)	(0.033)
CEO education		0.013***	0.011***
		(0.003)	(0.003)
CEO information		−0.022**	−0.038***
		(0.010)	(0.009)
Policy Regulation		−3.585***	−1.001
		(1.072)	(0.974)
Constant	0.043***	−0.221***	−0.269***
	(0.002)	(0.065)	(0.062)
Observation	877	877	877
R-Squared	0.021	0.138	0.223
Industry	No	No	YES
Year	No	No	YES

This table shows the robustness of the effects of corporate AI adoption rate on green innovation efficiency by narrowing the studying period in 2016 to 2020. Column (1) has neither control nor fixed variables; Column (2) contains control but no fixed variables; Column (3) contains both control and fixed variables. The robust standard errors are at firm-level. * stands for 10% significant level, ** and *** stands for 5% and 1% level, respectively.

AI adoption rate, thereby underscoring the robustness of the results and suggesting that the baseline model is valid after narrowing the study period to 2016–2020.

4.4.2. Alternative dependent variable

We further evaluate the robustness of our findings by substituting corporate GIE with the number of corporate green patents (Darendeli et al., 2022), which is an indicator of the natural logarithm of this variable. Data pertaining to green patents held by Chinese A-share listed energy companies are sourced from the CSMAR database. Following the research steps outlined in Section 4.2, the results displayed in Table 9 show uniformly positive and significant coefficients across Columns (1)–(3). These results suggest that the adoption of AI technology significantly increases the number of green patents held by energy companies, thereby reinforcing the validity and robustness of our primary findings.

5. Mechanism and heterogeneity test

5.1. ESG activities and executive myopia as moderating effects

5.1.1. ESG activities

In accordance with Hypothesis 2, we construct a proxy for ESG activities as follows. Regarding corporate environmental performance, the scarcity of publicly available information presents a challenge. Thus, we employ the number of environmental penalties levied against a company during a specific year, as disclosed on the website of the Institute for Public and Environmental Affairs, as an indicator of corporate environmental performance. The annual number of environmental penalties imposed on each listed company and its subsidiaries is aggregated to obtain the total number of environmental penalties imposed by the government during the year. A higher value for this indicator signifies that a company receives more environmental penalties during the year, which indicates worse environmental performance.

Table 9
Alternative dependent variable - Number of green patent.

	(1)	(2)	(3)
	Green Patent	Green Patent	Green Patent
AI	0.508***	0.269*	0.552***
	(0.146)	(0.157)	(0.124)
Asset		0.206***	0.196***
		(0.057)	(0.055)
Leverage		−0.865***	−1.575***
		(0.218)	(0.241)
ROA		0.988*	−0.494
		(0.599)	(0.640)
BM		−0.136***	−0.098***
		(0.031)	(0.033)
Tobin's Q		−0.080***	−0.049*
		(0.029)	(0.028)
TOP1		−0.207	−0.103
		(0.209)	(0.214)
Firm Age		0.047	0.042
		(0.111)	(0.119)
Cash Flow		−1.950***	−0.438
		(0.459)	(0.512)
CEO education		0.185***	0.168***
		(0.046)	(0.048)
CEO information		−0.369**	−0.653***
		(0.174)	(0.141)
Policy		−63.420***	−24.683*
		(14.225)	(13.876)
Constant	0.794***	−6.285***	−6.166***
	(0.035)	(1.152)	(0.985)
Observation	1208	1208	1208
Adjusted R-squared	0.012	0.186	0.290
Industry FE	No	No	YES
Year FE	No	No	YES

This table shows the robustness of the effects of corporate AI adoption rate on green innovation efficiency by replacing green innovation efficiency to number of green patents from 2010 to 2020. Column (1) has neither control nor fixed variables; Column (2) contains control but no fixed variables; Column (3) contains both control and fixed variables. The robust standard errors are at firm-level. * stands for 10% significant level, ** and *** stands for 5% and 1% level, respectively.

Moreover, corporate governance is a multidimensional variable. In this study, we perform a principal component analysis of several factors, including the proportion of independent directors, concurrent appointment of the chairperson and general manager (dummy variable), state ownership (dummy variable), and the establishment of a strategic committee. We select the first principal component as a proxy variable for corporate governance capacity. After standardization, a larger value for this indicator implies superior corporate governance capacity.

Given that CSR reporting is not mandatory, the quality of such reports varies significantly across companies. Some reports disclose excess ambiguous information, potentially serving as a means to whitewash a company's image. We perform a principal component analysis of several factors, including the annual amount of corporate social donations and the existence of policies protecting shareholders, creditors, employees, safe producers, suppliers, and consumers. We select the first principal component as a proxy for CSR performance. After standardization, a larger value of this indicator implies superior CSR performance.

By constructing the abovementioned corporate ESG indicators, we introduce ESG activities as a moderating variable in the mechanism analysis. Table 10 presents the results, with the interaction term between the AI adoption rate and ESG activities serving as our primary focus. The results in Column (1) suggest that, under the moderating effect of ESG activities, a higher degree of AI adoption corresponds to

Table 10
Mechanism Analysis.

	(1)	(2)
	GIE	GIE
AI	0.017	0.030***
	(0.012)	(0.007)
AI×ESG	0.013*	
	(0.007)	
AI×Myopia		−0.003*
		(0.002)
Asset	0.009***	0.009***
	(0.003)	(0.003)
Leverage	−0.080***	−0.079***
	(0.013)	(0.013)
ROA	−0.022	−0.020
	(0.035)	(0.035)
BM	−0.005***	−0.005***
	(0.002)	(0.002)
Tobin's Q	−0.004***	−0.004***
	(0.002)	(0.002)
TOP1	−0.006	−0.007
	(0.012)	(0.012)
Firm Age	0.0005	0.0003
	(0.006)	(0.006)
Cash Flow	−0.027	−0.029
	(0.028)	(0.028)
CEO Education	0.008***	0.008***
	(0.003)	(0.003)
CEO Information	−0.034***	−0.034***
	(0.007)	(0.007)
Policy	−1.119	−1.323*
	(0.738)	(0.722)
Constant	−0.243***	−0.241***
	(0.051)	(0.051)
Observation	1208	1208
Adjusted R-Squared	0.212	0.213
Industry FE	YES	YES
Year FE	YES	YES

This table reports the mechanism analysis by using corporate ESG and executive myopia as moderating variables. Column (1) shows the results of ESG as moderating variable in the analysis; Column (2) shows the results of executive myopia as moderating variable in the analysis. All control variables, and industry and year fixed effects are included. The robust standard errors are at firm-level. * stands for 10% significant level, ** and *** stands for 5% and 1% level, respectively.

greater corporate GIE. This implies that ESG activities can positively accelerate the effects of AI adoption on corporate GIE to a certain extent, providing evidence supporting Hypothesis 2 at the 10% level.

Our mechanism analysis further indicates that increasing investment in ESG activities can enhance the positive influence of AI adoption on corporate GIE. Consequently, we propose that energy companies should bolster their investment in ESG activities, thereby rendering a more pronounced promotional effect of AI adoption on corporate GIE. Governmental entities could also introduce a range of policies to stimulate energy companies' greater participation in ESG activities or alleviate the financial burden associated with such investments. These results provide a valuable addition to the existing literature by offering a comparative perspective on previous findings (Gu et al., 2021).

5.1.2. Executive myopia

Based on Hypothesis 3, we use a method similar to that used in constructing the AI adoption rate in Section 3.2 to create a managerial myopia index, which serves as another moderating variable in our mechanism analysis. The steps involved in the textual analysis are as follows: First, we use a lexicographical approach and the MD&A statements found in the yearly reports of 500 A-share listed companies to establish a set of seed words indicative of managerial myopia. Second, given that synonyms are often used in annual reports, we construct a multidimensional vector to distinguish similar words between vectors. Third, we use the CBOW model in Word2Vec to construct an annual financial report corpus. Finally, we determine the frequency of words related to managerial myopia, based on the context provided by the CBOW model. A higher value for the managerial myopia index implies that managers in a given company are more inclined to prioritize short-term goals over long-term development. The detailed seed words are provided in Appendix B.

The coefficient of the interaction term between the AI adoption rate and managerial myopia shown in Column (2) of Table 10 suggests that managerial myopia negatively moderates the association between the AI adoption rate and corporate GIE, indicating the statistical significance of Hypothesis 3. In other words, corporate executives who focus on short-term gains can diminish the positive effects of AI adoption on corporate GIE, thereby undermining a company's green innovation production performance. The benefits of adopting AI technologies for green innovation may be less pronounced in energy companies led by such executives. Thus, this approach offers a novel research perspective that enriches and extends the insights reported in previous studies (Adomako and Nguyen, 2023).

5.2. Heterogeneity test

5.2.1. Dual role of CEO and chairman

The consolidation or separation of the roles of CEO and chairperson of the board can influence the CEO's control over the company (Ballinger and Marcel, 2010). If the roles are separate, the CEO's decision-making capacity and strategic planning for the company's green innovation may lack the absolute leadership power inherent in the role of a CEO who is also the board chairperson. This can subsequently affect a company's green innovation development and future planning. As a result, we divide the firms into two groups: those in which the CEO and chairperson roles are held concurrently by the same individual, and those in which the two roles are held by different people. As shown in Columns (1) and (2) of Table 11, the effect of AI adoption on corporate GIE is significantly and positively associated with CEO and chairperson roles held by the same individual. However, this is not significant in firms in which roles are held by different people. This suggests that we lack sufficient evidence to assert whether the AI adoption rate has a positive or negative impact on corporate GIE in energy firms in which the CEO and chairperson roles are held by different individuals.

5.2.2. State-owned and non-state-owned companies

Given the unique characteristics of Chinese firms' ownership structures, we categorize energy companies as state-owned and non-state-owned for a heterogeneity analysis (Jara et al., 2022). The operational modes and manifestations of state-owned companies differ vastly from those of non-state-owned companies, making it worthwhile to examine their respective performances in relation to green innovation following AI technology integration (Peng et al., 2016). We conduct separate regression analyses for energy companies that were state-owned and otherwise. The results in Columns (3) and (4) of Table 11, show that the positive association between AI adoption and GIE is significant at the 1% level for state-owned energy companies. However, we lack sufficient statistical evidence to assert that this phenomenon is also applicable to non-state-owned energy companies. Thus, the positive effect of AI technology use on corporate GIE is evident only in state-owned energy companies.

5.2.3. Whether or not companies hold bank shares

We further divide the energy companies into two distinct groups: those that possess bank shares and those that do not (Knaup and Wagner, 2012). Column (5) of Table 11 shows the influence of companies holding bank shares on corporate GIE following the incorporation of AI technology. In contrast, Column (6) shows the results for companies that do not hold bank shares. The results in Column (6) show a significant positive correlation between AI adoption and corporate GIE; however, the results in Column (5) reveal no substantial evidence supporting the

Table 11
Heterogeneity Analysis.

	Dual position of CEO and Chairman		Ownership of corporations		Corporations hold bank share	
	(1)	(2)	(3)	(4)	(5)	(6)
	Dual positions	Not dual positions	State owned	Non-state owned	Bank share	No bank share
AI	0.037***	−0.060	0.040***	−0.013	0.057	0.031***
	(0.007)	(0.054)	(0.007)	(0.054)	(0.069)	(0.007)
Asset	0.008**	0.010	0.016***	0.006	0.013	0.008**
	(0.003)	(0.010)	(0.004)	(0.006)	(0.015)	(0.003)
Leverage	−0.078***	−0.072*	−0.085***	−0.049**	−0.006	−0.079***
	(0.014)	(0.040)	(0.017)	(0.021)	(0.065)	(0.013)
ROA	0.009	−0.083	−0.147**	0.041	0.508*	−0.043
	(0.039)	(0.075)	(0.059)	(0.042)	(0.265)	(0.036)
BM	−0.004**	−0.011**	−0.003	−0.005	−0.003	−0.004**
	(0.002)	(0.005)	(0.002)	(0.004)	(0.007)	(0.002)
Tobin's Q	−0.003*	−0.008	−0.003*	−0.002	−0.011	−0.002
	(0.002)	(0.005)	(0.002)	(0.003)	(0.014)	(0.002)
TOP1	−0.020*	0.113***	−0.009	0.013	−0.154*	0.006
	(0.012)	(0.042)	(0.017)	(0.020)	(0.087)	(0.012)
Firm Age	−0.005	−0.002	0.022**	−0.017*	−0.064	−0.009
	(0.007)	(0.023)	(0.010)	(0.009)	(0.060)	(0.007)
Cash Flow	−0.013	−0.144**	0.017	−0.044	−0.281	−0.031
	(0.031)	(0.064)	(0.038)	(0.042)	(0.190)	(0.028)
CEO education	0.007***	0.008	0.004	0.011**	−0.003	0.012***
	(0.003)	(0.009)	(0.003)	(0.005)	(0.012)	(0.003)
CEO information	−0.026***	−0.111***	−0.052***	−0.032***	−0.049**	−0.033***
	(0.008)	(0.024)	(0.008)	(0.012)	(0.023)	(0.009)
Policy	−1.650**	1.843	−0.335	−3.051**	1.400	−1.048
	(0.792)	(3.449)	(0.816)	(1.531)	(3.886)	(0.730)
Constant	−0.213***	−0.304**	−0.386***	−0.251***	0.059	−0.296***
	(0.055)	(0.141)	(0.075)	(0.094)	(0.297)	(0.052)
Observation	1035	148	756	452	94	1114
Adjusted R-Squared	0.240	0.419	0.346	0.244	0.379	0.273
Industry FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES

This table reports the heterogeneity analysis between AI adoption rate and corporate green innovation efficiency. Column (1) and (2) show the estimated results of whether companies have dual positions of CEO and chairman or not; Column (3) and (4) show the estimated results of whether companies are state-owned or not; Column (5) and (6) show the estimated results of whether companies hold bank share or not. All control and fixed variables are included. The robust standard errors are at firm-level. * stands for 10% significant level, ** and *** stands for 5% and 1% level, respectively.

effect of AI technology on green innovation in energy companies that hold bank shares. This disparity suggests that the influence of AI technology on a firm's green innovation varies between energy firms that hold bank shares and those that do not, indicating heterogeneity in this effect.

6. Conclusions and policy implications

6.1. Conclusions

The advent of AI has recently revolutionized the operational planning and production modes of publicly listed companies. This study focuses on 279 energy companies listed on China's A-share markets, using data from 2010 to 2020. We use textual analysis methods from the field of machine learning to construct an indicator of the AI adoption rate, which provides an objective and realistic depiction of the extent to which energy companies have embraced AI technology. We then apply an array of econometric models, including basic regression, generalized bifurcation, robustness testing, mechanism analysis, and heterogeneity analysis, to empirically examine the influence of the AI adoption rate on green innovation in energy companies. Based on our findings, we can draw conclusions and offer recommendations for both energy companies and policymakers. Emphasizing AI technology adoption within energy companies holds the promise of significantly bolstering green innovation output. This positive effect can be further amplified when energy companies increase their efforts toward ESG initiatives. However, a vigilant approach is essential among executives because the potentially deleterious moderating effects of short-termism on both AI technology and GIE warrant careful consideration.

Our conclusions are as follows. First, the baseline regression results indicate that AI adoption can significantly enhance energy companies' GIE. Second, the endogeneity test reveals that the current period of AI adoption positively affects corporate green innovation during subsequent periods. Thus, AI technology not only boosts the current period's green innovation knowledge output but also positively affects future green goal setting and implementation, providing valuable insights for long-term planning within energy companies. Third, our mechanism analysis reveals that a company's ESG performance can positively moderate the effects of AI adoption on corporate GIE. Further analysis from the perspective of internal executives indicates that executives who focus on short-term benefits negatively moderate the effect of AI adoption on corporate GIE, thereby diminishing its positive effects. Fourth, according to the heterogeneity test, the effect of AI adoption on corporate GIE is significantly positively correlated with firms in which the CEO also holds the chairperson position, that are state-owned, and that do not hold bank shares. Fifth, this study utilizes the environmental protection mandate issued by President Xi in 2017 as a temporal reference point and performs an empirical analysis using the generalized difference-in-difference model. The results indicate that even post-2017, AI adoption has continued to exert a significant positive influence on energy companies' GIE.

Based on the above findings, the main conclusions have several valuable implications for energy companies. First, in this era of heightened environmental consciousness, energy companies can strategically adopt AI technology to stimulate green innovation, thereby enhancing their environmental performance. Second, based on the mechanism analysis of ESG, we propose that energy companies should not only incorporate AI technology into their green innovation activities but also augment their investment in ESG activities to achieve their goals for sustainable development. Third, based on the findings of the heterogeneity test, energy companies should enhance the effectiveness of AI technology in relation to corporate green innovation and sustainable development by allowing the CEO to also hold the board chairperson position, appropriately nationalizing the company's capital, and reducing the company's bank shareholding when feasible.

This study has several important implications for executives. First, by investing more in AI technology, energy companies can increase their green patent output. Second, the mechanism analysis of executive myopia suggests that corporate executives should shift their focus away from immediate gains and toward long-term development. Although the benefits of adopting AI technology may not be immediately apparent, a lagged analysis of the robustness check indicates that introducing AI technology could stimulate green innovation output in the future.

6.2. Policy implications

This study provides valuable insights for policymakers. First, to encourage energy companies to better align with national environmental protection objectives, relevant governmental bodies could introduce incentives to motivate energy companies to enhance their green innovation output by adopting AI technology. This would facilitate the fulfillment of the government's environmental policy requirements for energy companies. Second, identifying lagged effects in the relationship between AI adoption rates and GIE has significant implications. Notably, the positive effects extend beyond the present and into the future. This insight serves as a source of inspiration for policymakers, suggesting that they should align their strategies with the United Nations' SDGs. The government can play a pivotal role by formulating policies that incentivize energy companies to leverage AI technology for the explicit purpose of fostering the creation of green patents, thereby contributing to the achievement of environmental objectives.

Inclusion and diversity

While citing references scientifically relevant for this work, we also actively worked to promote gender balance in our reference list. The author list of this paper includes contributors from the location where the research was conducted who participated in the data collection, design, analysis, and/or interpretation of the work.

CRediT authorship contribution statement

Zongrun Wang: Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Taiyu Zhang:** Writing – original draft, Formal analysis, Data curation. **Xiaohang Ren:** Writing – review & editing, Validation, Methodology, Conceptualization. **Yukun Shi:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Seed words of AI Adoption Rate in Chinese

Name	Chinese Seed Words	English Translation of Seed Words
Seed Words Set	采纳、应用、引入、开发、智能、AI、算法、自主、机器人、自动、识别、模拟、优化	adopt, application, introduce, develop, intelligence, AI, algorithm, autonomous, robot, automatic, identify, recognition, simulation, optimize
Expanded words Set	智能转型、数字智能、智能语言、机器学习、深度学习、人机协同、智能办公、智慧公司、智能决策、智能监督、智能产业、工业机器、数字化、自适应、智能化、智能技术、智慧财务、智能会计、记忆网络、目标检测、人脸识别、神经网络、自主决策、特征选择、语言处理、文本识别	intelligent transformation, digital intelligence, intelligence language, machine learning, deep learning, human-machine cooperation, smart office, intelligent company, intelligent decision, intelligent supervision, intelligent industry, industrial robot, digitalization, self-adaption, intelligent transformation, smart technology, smart finance, smart accounting, memory network, target detection, facial recognition, neural network, autonomous decision-making, feature selection, language processing, textual recognition

This table illustrates the Chinese seed words by using textual analysis to construct the index of AI adoption rate in energy companies. The English translation of seed words are also included.

Appendix B. Seed words of Executive Myopia in Chinese

Name	Chinese Seed Words	English Translation of Seed Words
Seed Words Set	天内、数月、年内、尽快、立刻、马上(直接表示)；契机、之际、压力、考验(间接表示)	within day(s), several months, within year(s), as soon as possible, immediately, soon (direct express); chance, when, pressure, challenge (indirect express)
Expanded words Set	日内、数天、随即、即刻、在即、最晚、最迟、关头、恰逢、来临之际、前夕、适逢、遇上、正逢、之时、难度、困境、严峻考验、双重压力、通胀压力、上涨压力、应尽快、尽早、早日、及早、时值、时机、到来之际、财务压力、环境压力、诸多困难、融资压力、还款压力	within a day, several days, presently, instantly, at present, the latest, late, moment, prior, right time, encounter, on time, at once, difficulty, serious challenge, double pressure, inflation pressure, upward pressure, as quickly as possible, as early as possible, sooner, in advance, before too late, on the spot, opportunity, approaching, financial pressure, environmental pressure, several difficulties, capitalize pressure, repayment pressure

This table illustrates the Chinese seed words by using textual analysis to construct the index of corporate executive myopia. The English translation of seed words are also included.

Appendix C. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eneco.2024.107499.

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Corrigendum to "AI adoption rate and corporate green innovation efficiency: Evidence from Chinese energy companies" [Energy Economics 132 (2024) 107499]

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The authors noted the format of eq. (2) has some missing parts due to operational error during formatting adjustment. The denominator of eq. (2) should be the natural logarithm of annual R&D expenditure plus one. The results of this article are not affected by the formatting error.

3.3. Measure of corporate GIE

Line 6 of the first paragraph should be: "We measure corporate GIE by the natural logarithm of total number of green patents, utility models, and design patents plus one divided by the natural logarithm of annual R&D expenditure plus one."

The correct formula we use to measure GIE is as follows:

$$GIE = \frac{ln[(green\ patents) + (utility\ models) + (design\ patents) + 1]}{ln[(Annual\ R\&D\ expenditure) + 1]} \tag{2}$$

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